Prove the fact, not your ID.

Zero-knowledge credentials from the IC chip in Japan's My Number Card, verified in any browser and on Solana.

Yu Ayato



THE PROBLEM · WORLDWIDE

ID documents are leaking everywhere.

US & CANADA · SEP 2026

150M+

records from an ID-verification vendor: driver's licences, passport numbers

JAPAN · SEP 2026

1.6M

ID images from a car-sharing service's sign-ups

GLOBAL · OCT 2025

~70k

government ID photos sent for age-check appeals on a chat platform

US · JUL 2025

13k

selfies and photo IDs uploaded to verify accounts on a dating-safety app

AUSTRALIA · MAR 2023

7.9M

driver's licence numbers (and 53k passport numbers) from a consumer lender

Every one was a copy someone was asked to upload.

Sources: TechCrunch (Sep 2026; 150M+ as reported, company confirmed licence and passport data), company disclosures and NHK (Japan), TechCrunch / NBC (Oct 2025), Reuters via Fortune (Jul 2025), Latitude Financial (Mar 2023).

THE PROBLEM · IT IS ABOUT TO GROW

Laws now demand age and ID checks — and most checks still mean uploading your ID.

UK · JUL 2025

Online Safety Act: "highly effective" age checks for adult sites

AUSTRALIA · DEC 2025

Minimum age 16 for social media accounts

US · 2026

About half of the states mandate some form of age gating

JAPAN · APR 2027

Remote KYC moves from photo uploads to IC-chip reading

More checks × stored copies
= more breaches.

$4.44M

average cost of one data breach, worldwide (2025)

Sources: Ofcom; eSafety Commissioner; Jumio age-verification law snapshot (2026); JAFIC (NPA); IBM Cost of a Data Breach Report 2025.

CLOSE-UP · JAPAN, SEPTEMBER 2026

A car-sharing service lost its members' ID cards.

1.6M

images of driver's licences and other ID documents uploaded at sign-up

6.6M

member accounts: name, address, date of birth, licence data

30.6M

people hit by leaks at Japanese listed companies in 2025 alone

Sources: company disclosures (Sep 2026), NHK (Oct 2026), Tokyo Shoko Research (Jan 2026). Police are reissuing affected licences.

Data you never collect cannot leak.

TODAY

`collect → store → leak`

Every service keeps a copy of your ID.

WITH P!PON

`verify → forget`

The service checks a proof and keeps nothing.

This is the turning point: services will stop holding personal data wherever they can.

WHY START IN JAPAN

Japan is moving identity checks from photos to the chip.

Apr 2027

Remote KYC under the anti–money-laundering law drops photo uploads in favour of reading the IC chip.

104.5M

My Number Cards held (84.5% of the population). Each chip can carry a certificate signed by the national authority, J-LIS.

Sources: National Police Agency (JAFIC) Q&A on the amended ordinance; Ministry of Internal Affairs and Communications, cards held at end of Sep 2026.

IC chips solve authenticity.
Zero-knowledge proofs solve privacy.

My Number Card

Certificate signed by J-LIS. Signs a fresh challenge.



Phone

Checks the signatures inside a zero-knowledge proof.



Website or Solana

Verifies the proof. Gets a yes.

The service learns

Over 20 · lives in a prefecture · hasn't claimed before

It never sees

Name · date of birth · address · ID image · My Number

WHO IT IS FOR

Every service that asks for an ID only to learn one fact.

Online stores
Alcohol, tobacco, adult content

Sign-up checks
"A real, unique person"

One per person
Drops, airdrops, giveaways

DAO votes
Sybil-resistant governance

AI agents
Acting for a verified human

Not legal KYC
It's for everything else

WHY SOLANA

A shared "one per person" registry that no company has to own.

~144k compute units: Groth16 verify (alt_bn128), nullifier and NFT mint in **one transaction**

One per person, no list of people

Each card leaves a per-gate nullifier in a PDA. A second claim fails on-chain.

A trust list anyone can read

Gates hold the trusted J-LIS CA keys on-chain. Our Web2 store reads the same list: no API key, no vendor.

Next: attestations and agents

Solana Attestation Service for reusable credentials; x402 so an AI agent can show a verified human stands behind it.

Live on devnet: zk_gate program with real-card and demo gates · see the technical demo

ROADMAP

From one card in Japan to every chip, for people and their agents.

NOW · OCT 2026

Working sample

My Number Card: age + prefecture + one per person

zk_gate on Solana devnet; browser verifier for Web2

iPhone on-device proof (9 s); Mac + USB reader

Tested end to end on a real card

NEXT · BY Q1 2027

Product for stores

iPhone app in TestFlight, then Android

Drop-in checkout SDK for online stores

More facts: age lines, city, recent certificate

Solana Attestation Service; multi-party setup, audit, mainnet

AI AGENTS · 2027

A human behind the agent

Delegate a scoped, revocable "verified human, 20+" credential to your agent

Proof attached to x402 payments for age-restricted goods

Per-human limits via nullifiers: one person, N agents

Tools for agent wallets and MCP

GLOBAL · 2027+

Every ID chip

ePassports to the ICAO 9303 standard: chip data signed by each country's CA

Other national eID cards with signed chips

Same proof, same Solana registry, any country

Revocation through certified operators

STATUS

Working today on devnet, end to end with a real My Number Card.

Next: stores, AI agents, then every ID chip — see the roadmap.

Stop collecting IDs.
Prove the fact.



TECHNICAL DEMO · ARCHITECTURE

No issuer. The national CA is the root of trust; the phone does the proving.

Layer	Built with
Circuit	Circom · Groth16 · 1.49M constraints
Prover	mopro + rapidsnark on iPhone · Rust CLI on Mac
Card I/O	Core NFC (iPhone) · CryptoTokenKit / PC/SC (Mac)
On-chain	Anchor program zk_gate · alt_bn128 · Metaplex Core
Web verifier	snarkjs in the browser · CA trust list read from Solana

INSIDE THE PROOF

Four checks in private. This sample reveals two facts.

1. J-LIS CA signature over the certificate (RSA-2048, SHA-256)

2. The card's signature over a challenge bound to the wallet or browser session

3. Date of birth + N years ≤ today

4. Address starts with the chosen prefecture

Revealed in this sample

Age ≥ 20 · Prefecture

Plus the plumbing: CA key hash, a per-service nullifier, challenge, date, scope.

Room to grow — the certificate also holds name, date of birth, sex and full address, so the same pattern can prove (not built yet):

- Other age lines: 18+, 65+, under 25
- City or ward
- Sex, only where a service must know
- Name matches the account (hash, not the name)
- Certificate issued within N days

FLOW 1 · WEB2 · ONLINE STORE SELLING ALCOHOL

How the store learns "20+" without seeing your ID

STORE

1 · Checkout asks

Shopper taps "Verify age with p!pon". The store makes a one-time session ID and shows it as a QR code (a link on mobile).

PHONE

2 · Request opens

The p!pon app shows who asks and what: "Kura Online Store · Age 20+".

CARD

3 · Tap the card

After the signing password, the chip signs SHA-256(session ID ‖ nonce) and gives its J-LIS-signed certificate.

PHONE

4 · Prove on device

Groth16 proof (~9 s): CA signature, card signature, age ≥ 20, prefecture. The certificate never leaves the phone.

STORE'S BROWSER

5 · Verify, ~200 ms

Proof valid · same session (no replay) · CA in the trust list on Solana · dated today · 20+ · scoped to this store.

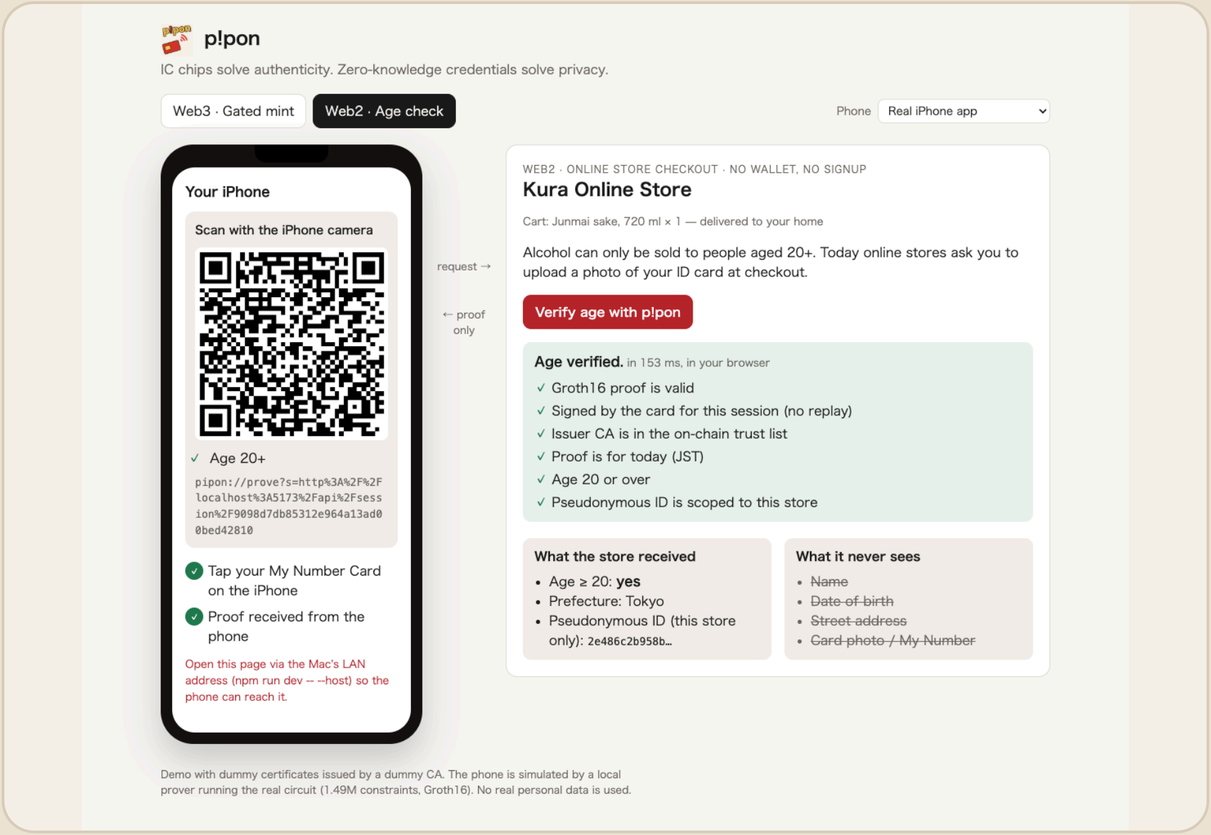
Store keeps: age 20+ = yes, prefecture, a store-only pseudonymous ID.

Never receives: name, date of birth, address, ID image, My Number.

Once per account or order, from home — not a card scan at a shop counter.

DEMO 1 · WEB2 · ONLINE STORE CHECKOUT

No wallet, no signup: the store gets a yes.



An online store selling sake: age 20+ needed at checkout

Verified in the browser in ~200 ms; issuer CA checked against the trust list on Solana

Store learns: 20+, prefecture, a pseudonymous ID for this store only

FLOW 2 · WEB3 · ONE-PER-PERSON MINT

How Solana enforces "20+, one per person"

WALLET

1 · Open the gate

The gate account on Solana holds the rules: minimum age, prefectures, trusted CA keys, scope.

PHONE

2 · Bound to the wallet

The challenge is SHA-256(wallet address ‖ nonce), so the proof only works for this wallet.

CARD + PHONE

3 · Tap and prove

The card signs the challenge; the phone makes the same Groth16 proof as in Web2, with the gate's scope.

SOLANA · zk_gate

4 · Verify on-chain

Groth16 via alt_bn128 syscalls; CA, age, prefecture, date and the signer's wallet checked against the gate.

SOLANA · zk_gate

5 · Mint or reject

New nullifier → PDA created, Core NFT minted, same transaction (~144k CU). Nullifier exists → "Already claimed".

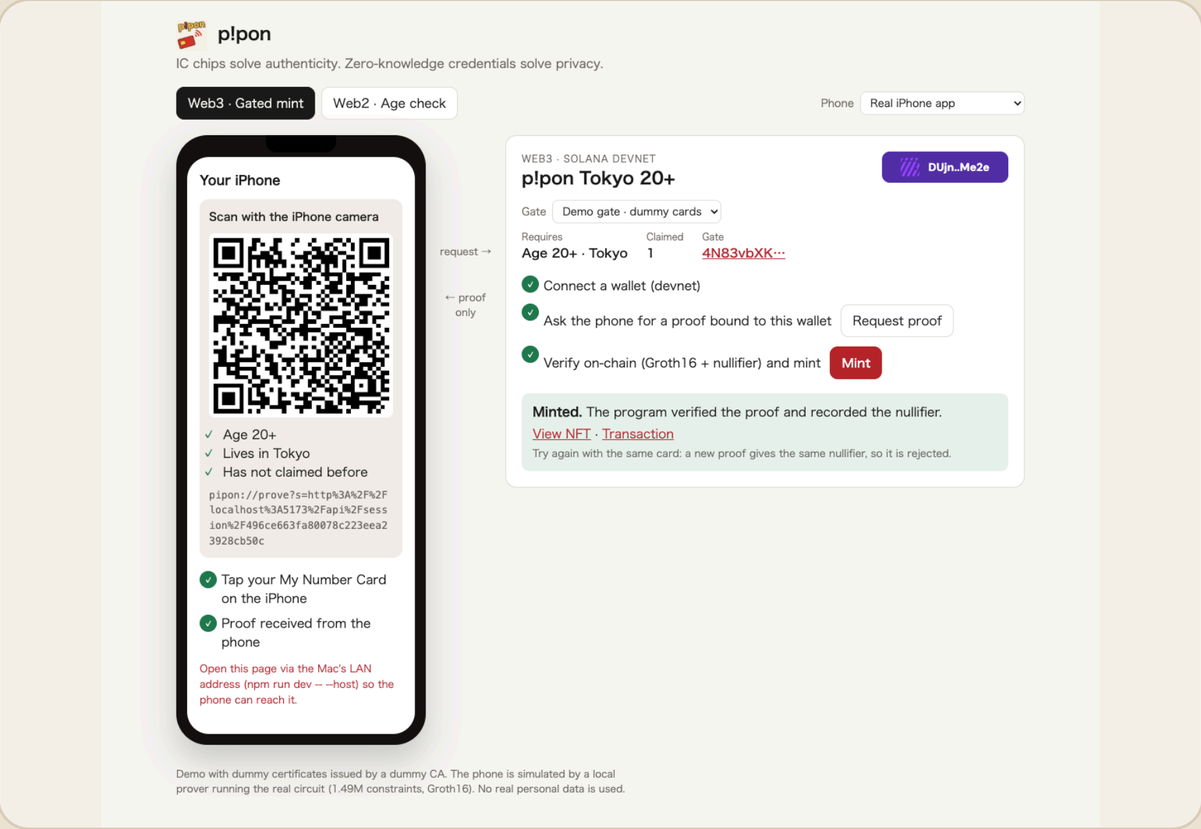
Public on-chain: the proof and its values — CA key hash, nullifier, date, age threshold, prefecture code.

Never on-chain: name, date of birth, address, ID image, My Number.

The nullifier is per card and per gate: another gate gets a different one, so wallets can't be linked across services.

DEMO 2 · WEB3 GATED MINT

One transaction: verify, record, mint.



Proof bound to the wallet

Same card again → **Already claimed**

Card from another prefecture → **rejected by the gate**

TESTED ON A REAL MY NUMBER CARD

Same flow, real J-LIS certificate.

3.5 s

proof on a Mac mini, card in a USB reader

9 s

on-device proof on an iPhone 15 Pro Max

2 / 2

claim verified on devnet · second claim from a new wallet rejected

Devnet test gate (scope 3), 2026-10-07 · tx c623tn…zvYv7 · no personal data shown or stored

LIMITS AND NEXT

Honest about what's left.

Limits today

No revocation check yet: revocation lists are only for certified verifiers.

Single-contributor trusted setup: demo only.

How the roadmap closes them

Revocation through a certified platform operator.

Multi-party trusted setup and an audit before mainnet.

iPhone NFC in TestFlight (paid signing team pending).